NATHISHWAR C

AIEngineer | LLMApplications | RAGSystems | FastAPIBackend
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Portfolio | Linked in | Github

SUMMARY

AI Engineering student specializing in building production-ready AI systems, LLM-powered applications, and scalable backend services. Experienced in developing adaptive learning platforms, RAG pipelines, and AI-driven recommendation systems using FastAPI, PostgreSQL, and vector databases. Creator of KnowGrow AI — an intelligent learning platform that personalizes education using adaptive spaced repetition algorithms and LLM tutoring.

SKILLS SUMMARY

Programming: Python, SQL, JavaScript
Machine Learning: Scikit-learn, Feature Engineering, Model Evaluation
Deep Learning: PyTorch, Neural Networks
Generative AI: LangChain, RAG, HuggingFace Transformers
Backend: FastAPI, REST APIs
Databases: PostgreSQL, Redis, FAISS Vector DB
DevOps: Docker, Linux, CI/CD
Tools: Git, GitHub, Postman, VS Code

PROJECTS

KnowGrow AI — Adaptive Learning Intelligence Platform

Designed and developed an AI-powered adaptive learning platform that generates personalized courses and optimizes revision using intelligent learning algorithms.

•Builtan AI recommendation engine that analyzes user learning patterns to generate personalized study paths

•Implemented confidence-based spaced repetition algorithm to optimize knowledge retention and revision scheduling

•Integrated LLM APIs to generate learning content, quizzes, and tutoring assistance •Developed scalable backend using FastAPI, PostgreSQL, Redis, and Docker •Designed RAG pipeline for contextual knowledge retrieval

AIKnowledge Assistant (RAG-based)

Developed an AI assistant capable of retrieving knowledge from documents and generating contextual responses using retrieval-augmented generation.

•Builtdocument ingestion pipeline with embeddings and vector indexing

•Implemented semantic search using FAISS vector database

•Integrated HuggingFace embedding models with LLM response generation

•Developed scalable API backend using FastAPI for chatbot interaction

•BuiltRAG system capable of retrieving contextual answers from 500+ documents using FAISS

WORK EXPERIENCE

AI/ML Intern — Google Developer Student Club (GDSC)
Remote | Oct 2024 – Dec 2024

•Developed machine learning models for predictive analytics using Scikit-learn including Logistic Regression and Decision Trees

•Implemented data preprocessing and feature engineering pipelines for structured datasets

•Built end-to-end ML workflows including training, evaluation, and prediction pipelines

•Collaborated in team-based AI development environment using Git and modern ML practices

EDUCATION

B.Tech Artificial IntelligenceandDataScience — **9.0CGPA (Till *5th Sem)**
Dhanalakshmi SrinivasanEngineeringCollege — Aug 2023 - May 2027

Marutham matric.Hr.Sec.School — **89.8%**
Bio Mathematics — Jun 2021 - May 2023

GITHUB HIGHLIGHTS

• Developed 10+ AI and backend projects including LLM applications, RAG systems, and scalable FastAPI services.
• Built LLM-powered applications using LangChain and RAG architecture
• Implemented scalable APIs using FastAPI and PostgreSQL

CERTIFICATES

- Programming in Python — Simplilearn
- Machine Learning — Unified Mentor
- Problem Solving — HackerRank
- AI Foundations — LinkedIn Learning